UNIT- I

Q. What is Discrete Mathematics?

Ans: Discrete mathematics is the part of mathematics devoted to the study of discrete objects.

Discrete means "distinct or unconnected elements".

Discrete object is something that is countable.

Examples: ① The Integers or natural numbers.

② The rational numbers ③ Finite sets

④ Functions from $\{1, 2, \ldots, n\} \rightarrow \{0, 1\}$

⑤ People, chairs, tables, balls, ...

Q. Why study Discrete Mathematics?

→ ① It develops your mathematical thinking.

② Improves your problem solving ability.

③ Foundation for many courses in Computer Science/Eng.

- Data structures, algorithms (Graph, Computability).
- Artificial Inteligence (logic, graph, Automata)
- Databases (Relations, logic)
- Computer N/W's (Graphs).
- Compiler design, formal languages, automata theory, Computer security, and operating systems.

④ Foundation for a new discipline: Formal method for CS/Eng.

- proving correctness of programs.
- proving properties of S/W systems: Deadlock-free.
- Verifying protocols: ISDN protocol, security protocol.
- Finding bugs in μp's. (used by Intel, IBM, Motorola)
- Verifying configurations of N/W systems – Firewalls.

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The phrase propositional logic is composed of two words:
- proposition
- Logic

↳ The area of logic that deals with propositions.
↳ also called propositional calculus.

Q. What is logic?

→ • Logic is the study of correct reasoning.
It helps us to understand and reason about different mathematical statements. The Rules of logic gives precise meaning to " Stmts.
• With rules of logic, we would be able to think about mathematical statements and finally we would be able to prove or disprove those mathematical statements precisely.

Use of logic: ① In Mathematics
↳ to prove theorems.
② In CS: to prove that programs do what they are supposed to do.

Logic focuses on the relationship among statements.

For example: ①
My watch is digital
All digital devices run on batteries.
Therefore, My watch runs on batteries.

Note that Logic is not concerned with the truth of the first two statements. But if they were true, then the inference is true.

② For every positive integer n, the sum of positive integers not exceeding n is $\frac{n(n+1)}{2}$.

purpose of logic is to construct valid arguments (or proofs).
Once we prove a mathematical statement is TRUE then we call it as a Theorem. and this is the basis of whole mathematics.

purpose of logic is to distinguish b/w valid and Invalid mathematical arguments.

Major goal of this Course Subject: how to understand & how to construct correct mathematical arguments.

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Q. What is proposition? → Statement

→ A proposition is a declarative sentence that is either true or false, but not both. The basic building blocks of logic. For example, "My name is Ramu" is a declarative statement. but "What is your name?" is not a declarative statement.

Examples :- ① Delhi is the Capital of India.
② $1+1=2$
③ 5 is a prime number.
④ $2+2=3$

propositions 1, 2, 3 are true, where as 4 is false.

Sentences which are not propositions:
(1) $x+y=z$
(2) How beautiful are you?
(3) Read this carefully.
④ What a beautiful Morning!
⑤ Do your homework.
⑥ $x+2=3$
⑦ What time is it?

propositional (Statement) Variable :-

A Variable that represents a proposition. Denoted by p, q, r, s. The truth or falsity of a proposition is called its truth-value. These two values 'true' and 'false' are denoted by the symbols T and F respectively. Sometimes these are also denoted by the symbols 1 and 0 respectively.
Truth value : True or False.

Compound proposition : (~~Control~~ Connectives)

A Compound proposition are formed by combining more than one propositions using logical operators. (or)
A proposition constructed by combining propositions using logical operators.
↳ operators used to combine propositions.
Compound proposition is also named as connectives.

Example: ① Rama is a boy (and) sita is a girl.
proposition 1 — Joining two propositions with logical connective — proposition 2

Q. Why do we need Compound propositions?

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Example ②: P: Einstein is a genius.

Q: It is not the case that Einstein is a genius.

Modifying the statement using negation.

There are Five basic connectives namely negation, Conjunction, disJunction, Conditional and biconditional, exclusive-or.

① Negation :- Let p be a proposition. The negation of P, denoted by $\neg p$, is the statement "It is not the case that P". The proposition $\neg p$ is read as "not p". [also named as $\sim$]

Example ①: P: Einstein is a genius.

$\neg P$: It is not the case that Einstein is a genius.

(or)

It is false that Einstein is a genius.

(or)

Einstein is not a genius.

Example 2: Q. Find the negation the proposition "Hyderabad is a city" and express this in Simple English.

A: The negation is "It is not the case that Hyderabad is a city.

(or)

Hyderabad is not a city.

Truth table:

p	$\neg p$
T	F
F	T

Truth table: a table displaying the truth values of propositions.

Types of Connectives (or) Compound propositions:-

1) Negation (not)
2) Conjunction (and)
3) Disjunction (or)
   ④ Exclusive-or ($\oplus$)
4) Conditional (if..then)
5) Bi Conditional (if and only if)

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② Conjunction :-

Let p and q be propositions. The conjunction of p and q, denoted by $p \wedge q$, is the proposition of p and q.

The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Truth Table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Note: The word "but" sometimes is used instead of "and" in conjunction.

Example: ① The Sun is shining, but it is raining.

is another way of saying "The sun is shining and it is raining".

② p: Jack went up the hill

q: Jill went up the hill.

$p \wedge q$: Jack and Jill went up the hill.

③ Disjunction :- Let p and q be propositions. The Disjunction of p and q, denoted by $p \vee q$, is the proposition "p or q".

The disjunction $p \vee q$ is false when both p and q are false and is true otherwise. Disjunction is also known as inclusive-or.

Truth Table :-

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

A disjunction is true when atleast one of the two propositions is true.

Types: Inclusive-or and Exclusive-or

Examples: ① I shall goto Market or Cinema.

② There is something wrong with the bulb or wiring.

③ It is raining or it is cold.

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(iii) Exclusive-or :-

Let p and q be propositions. The exclusive-or of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

Truth Table :-

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

(iv) Conditional or If.. then Statement :- (Implication)

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p, then q".

The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise.

In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called Conclusion (or consequence).

The Truth table for the conditional statement $p \rightarrow q$:-

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The statement $p \rightarrow q$ is called a conditional statement because $p \rightarrow q$ asserts that q is true on the condition that p holds. The conditional statement is also called an implication.

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The proposition $p \to q$ may be expressed as:

(1) "if p, then q"
(2) "if p, q"
(3) "p is sufficient for q"
(4) "q if p"
(5) "q when p"
(6) "a necessary condition for p is q"
(7) "q unless $\neg p$"
(8) "p implies q"
(9) "p only if q"
(10) "a sufficient condition for q is p"
(11) "q whenever p"
(12) "q is necessary for p"
(13) "q follows from p"

Example: ① If triangle ABC is equivalent, then it is isoScales.

② p: it is hot
q: $2+3=5$
$p \to q$: If it is hot, then $2+3=5$.

③ If you get 100% on the final (EXAM), then you will get an A. (grade).

④ If it is sunny tomorrow, you may go swimming.

Q. Write the following statements in symbolic form:

ⓐ If either Jerry takes Calculus or Ken takes Sociology, then Larry will take English.

ⓑ The crop will be destroyed if there is a flood.

Ans: ⓐ We denote the statement as

J: Jerry takes Calculus
K: Ken takes sociology
L: Larry takes English

These stmts can be symbolized as $(J \vee K) \to L$

ⓑ C: The crop will be destroyed
F: There is a Flood.
The statement can be symbolized as: $F \to C$

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There are three related Conditional Statements.

Implication $P \rightarrow q$ if P, then q P is sufficient for q q is necessary for P	**Converse** $q \rightarrow P$ If q, then P q is sufficient for P P is necessary for q.
Inverse/opposite $(\neg P) \rightarrow (\neg q)$ if not P, then not q (equivalent to the Converse)	Contrapositive $(\neg q) \rightarrow (\neg P)$ If not q, then not P (equivalent to the Implication)

Equivalent Statement :- When two Compound Statements always have the same truth value is called equivalent statement.

(a) Implication and its Contrapositive are equivalent.
(b) Converse and the inverse are also equivalent.
(c) Neither Converse nor Inverse is not equivalent to Implication.

Q(1): State (What are) the Contrapositive, Converse and inverse of the Conditional Statement.

"~~If the home team wins, then it is raining~~"
"If it is raining, then the home team wins".

(1) Contrapositive: $\neg P \quad (\neg q) \rightarrow (\neg P) \quad q$
"If the home team does not win, then it is not raining".

(2) Converse: $q \rightarrow P$
"If the home team wins, then it is raining".

(3) Inverse :- $\neg P \rightarrow (\neg q)$
"If it is not raining, then the home team does not win".

Q2. If triangle ABC is a right angle, then $|AB|^2 + |BC|^2 = |AC|^2$.

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(5) Biconditional statement :-

Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ (or) $p \rightleftarrows q$, read as "p if and only if q" or "p iff q". The statement $p \leftrightarrow q$ is true whenever both p and q have the same truth values and is false otherwise. It is also called bi-implication.

$p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

Truth table :-

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

The proposition $p \leftrightarrow q$ may be expressed as:

a) "p if and only if q"
b) "p iff q"
c) "if p then q, and conversely"
d) "p is necessary and sufficient for q"

Example :- Let p : You can take the flight and
q : You buy a ticket then $p \leftrightarrow q$ is the statement
"You can take the flight if and only if you buy a ticket".

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Well-formed formula (wff):-

A statement formula is not a statement; However, a statement can be obtained from it by replacing the variables by statements.

A Statement formula is an expression which is a string consisting of variables, parentheses, and connective symbols. Not every ~~symbol~~ string of these symbols is a formula. We shall now give a recursive definition of a statement formula, often called a well-formed formula.

A well-formed formula can be generated by the following rules:

1. A statement variable standing alone is a wff.
2. If $\mathcal{A}$ is a wff, then $\neg A$ is a wff.
3. If A and B are wff's, then $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, $(A \leftrightarrow B)$ are wff's.
4. A string of symbols containing the statement variables, connectives, and parenthesis is a wff, iff it can be obtained by finitely many applications of the rules 1, 2, and 3.

The following are wff's:-

$((\neg P) \vee Q)$, $\neg(P \wedge Q)$, $\neg(P \vee Q)$, $(P \rightarrow (P \vee Q))$, $((P \wedge Q) \vee R)$

$(P \rightarrow (Q \rightarrow R))$, $((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightleftarrows (P \rightarrow R)$.

The following are not wff's:

1. $\neg P \wedge Q$. obviously P and Q are wff's. A wff would be either $(\neg P \wedge Q)$ or $\neg(P \wedge Q)$
2. $(P \rightarrow Q) \rightarrow (\wedge Q)$. This is not a wff.
3. $(P \rightarrow Q$. parenthesis is missing.

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Precedence of logical operators :-

operator	precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Examples :-

$\neg p \wedge q$ is the conjunction of $\neg p$ and q, i.e $(\neg p) \wedge q$.

$p \wedge q \vee r$ means $(p \wedge q) \vee r$

$p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$

Construction of Truth Table :-

Q. Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$

Sol: Truth table of $(p \vee \neg q) \rightarrow (p \wedge q)$

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Rows: Need a row for every possible combination of values for the atomic propositions.

Columns: Need a column for the compound proposition (usually at far right)

- Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
- This includes the atomic propositions.

Q. How many rows are there in a truth table with n propositional variables?

Ans: With n propositional variables, we can construct 2^n distinct.

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Applications of propositional logic :-

1. Translating English sentences to propositional Logic.
2. System specifications.
3. Boolean Searches.
4. Logic puzzles.
5. Logic ~~circuits~~ and Bit operations.

1. Translating English Sentences :-

There are many reasons to translate English sentences into expressions involving propositional variables and logical Connectives. In perticular, English is often ambiguous. Translating sentences into compound statements removes ambiguity.

Steps: ① Identify atomic propositions and represent Using propositional variables.

② Determine appropriate logical Connectives.

Example ①: Translate the following English sentences into a logical expression.

(a) you can access the Internet from campus only if you are a Computer Science major or you are not a freshman.

Ans: a : you can access the Internet from Campus

c : you are a Computer Science major

f : you are a freshman.

only if is one way conditional statement.

Then the sentence can be translated to

$$a \rightarrow (c \vee \neg f)$$

(b) you cannot ride the rollercoaster if you are under 4 feet tall unless you are older than 16 years old.

P : you can ride the roller coaster

q : you are older than 16 years

r : you are under 4 feet tall

Then the sentence can be represented as

$$(r \wedge \neg q) \rightarrow (\neg p)$$

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② System Specifications:-

Translating sentences in natural language (such as English) into logical expressions is an essential part of specifying both hardware and software systems.
System and software engineers take requirements in natural language and produce precise and unambiguous specifications that can be used as the basis for system development.

Example: (a) Express the specification "The automated reply cannot be sent when the file system is full" using logical connectives.

Sol: Let P: The automated reply can be sent
q: The file system is full.
Specification can be represented by $q \rightarrow \neg p$.

Consistent System Specifications:- A list of propositions is consistent if it is possible to assign truth values to the proposition variables so that each proposition is true.

Example: Determine whether these system specifications are consistent.
"The diagnostic message is stored in the buffer or it is retransmitted."
"The diagnostic message is not stored in the buffer"
"If the diagnostic message is stored in the buffer, then it is retransmitted".

Sol: First, we express them using logical expressions.
Let P: The diagnostic message is stored in the buffer
q: The diagnostic message is retransmitted.
The specifications can be written as $p \vee q$, $\neg p$ and $p \rightarrow q$.
Because we want $p \vee q$ to be true but p must be false and q must be true.
Because $p \rightarrow q$ is true when p is false and q is true.
So, we conclude that these specifications are consistent.
because they are all true when p is false and q is true.

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③ Boolean Searches :-

Logical connectives are used extensively in searches of large collections of information, such as indexes of Web pages. Because these searches employ techniques from propositional logic, In Boolean Searches, the connective AND is used to match records that contain both of two search terms, the connective OR is used to match one or both of two search terms, and the connective NOT (sometimes written as AND NOT) is used to exclude a particular search term.

Example: Web page Searching. Most Web Search engines support Boolean searching techniques, usually can help find Web pages about particular subjects.
For instance, Using Boolean Searching to find Web pages about

① Universities in New Mexico.
↓
We can look for pages matching NEW AND MEXICO AND UNIVERSITIES.

② To find pages that deal with Universties in New Mexico or Arizona.
We can search for pages matching, (NEW AND MEXICO OR ARIZONA) AND UNIVERSITIES.

③ Universities in Mexico (and not New Mexico)
↳ (MEXICO AND UNIVERSITIES) NOT NEW. (or)
MEXICO UNIVERSITIES – NEW.
↳ NOT replaced by – (minus).

④ Logic puzzles :-

puzzles that can be solved using logical reasoning are known as logic puzzles. Solving logic puzzles is an excellent way to practice working with the rules of logic. Computer programs designed to carry out logical reasoning often use well-known logic puzzles to illustrate their capabilities.

Example: Raymond Smullyan, An island has two kinds of inhabitants, knights, who always tell the truth, and knaves, who always lie.
You go to the island and meet A and B.
- A says "B is a knight"
- B says "The two of us are of opposite types".

Example: What are the types of A and B?

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Sol: Let p and q be the statements that A is a knight and B is a knight, respectively. So, then $\neg p$ represents the proposition that A is a knave and $\neg q$ that B is a knave.

- If A is a knight, then p is true. Since knight tells the truth, q must also be true. Then $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true, but it is not. So, A is not a knight and therefore $\neg p$ must be true.
- If A is a knave, then B must not be a knight since knaves always lie. So, then both $\neg p$ and $\neg q$ hold since both are knaves.

(5) Logic and Bit operations:-

Computers represent information using bits. A bit is a symbol with two possible values, namely, 0 (zero) and 1 (one). bit represents a binary digit. i.e. zeros and ones are the digits used in binary representation of numbers.

A bit can be used to represent a truth value, because there are two truth values, namely true and false.

True (T) — 1

false (F) — 0

A variable is called a Boolean variable if its value is either true or false. Boolean variable can be represented using a bit. Computer bit operations correspond to the logical connectives. Replace true by a one and false by a zero in the truth tables for the operators $\wedge$, $\vee$, $\oplus$.

OR, AND, and XOR for the operators $\vee$, $\wedge$, and $\oplus$, as is done in various programming languages.

Truth table for the bit operators OR, AND, and XOR

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

A bit string is a sequence of zero or more bits. The length of a string is the number of bits in the string.

Example: 101010011 is a bit string of length nine.

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Example: Find the bitwise OR, bitwise AND, bitwise XOR of the bit strings: 01 1011 0110 and 11 0001 1101.

bitwise OR

```
01 1011 0110
11 0001 1101
-------------
11 1011 1111
```

bitwise AND

```
01 1011 0110
11 0001 1101
-------------
01 0001 0100
```

bitwise XOR

```
01 1011 0110
11 0001.1101
-------------
10 1010 1011
```

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Propositional Equivalences:-

① **Tautology**:- A Compound proposition that is always true. is called "Tautology".

Example: $P \vee \neg P$

P	¬P	P∨¬P
T	F	T
F	T	T

② **Contradiction**:- A Compound proposition that is always false is called a "Contradiction".

Example: $P \wedge \neg P$

P	¬P	P∧¬P
T	F	F
F	T	F

③ **Contingency**:- A Compound proposition that is neither a tautology nor a contradiction is called a contingency.

Example: $P \rightarrow q$, $P \leftrightarrow q$.

P	q	P→q	P↔q
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

Logical Equivalences:-

Compound propositions that have the same truth values in all possible cases are called "logically equivalent". The Compound propositions p and q are called logically equivalent if $P \leftrightarrow q$ is a tautology.

The notation $P \equiv q$ denotes that p and q are logically equivalent. The symbol $\Leftrightarrow$ is sometimes used instead of $\equiv$ to denote logical equivalence.

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Logical equivalences:-

Table 1

	Equivalence	Name
①	$P \wedge T \equiv P$ $P \vee F \equiv P$	Identity laws
②	$P \vee T \equiv T$ $P \wedge F \equiv F$	Domination laws
③	$P \vee P \equiv P$ $P \wedge P \equiv P$	Idempotent laws
④	$\neg(\neg P) \equiv P$	Double negation law
⑤	$P \vee Q \equiv Q \vee P$ $P \wedge Q \equiv Q \wedge P$	Commutative Laws
⑥	$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$ $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$	Associative Laws
⑦	$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$	Distributive Laws
⑧	$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$	De Morgan's Laws
⑨	$P \vee (P \wedge Q) \equiv P$ $P \wedge (P \vee Q) \equiv P$	Absorption Laws
⑩	$P \vee \neg P \equiv T$ $P \wedge \neg P \equiv F$	Negation Laws

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Table 2 : Logical equivalences involving conditional statements.

$$p \to q \equiv \neg p \vee q$$
$$p \to q \equiv \neg q \to \neg p$$
$$p \vee q \equiv \neg p \to q$$
$$p \wedge q \equiv \neg(q \to \neg p)$$
$$\neg(p \to q) \equiv p \wedge \neg q$$
$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$
$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$
$$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$$
$$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$$

Table 3: Logical equivalences involving biconditionals

$$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$$
$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$
$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$
$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Q. Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

$\therefore \neg(p \vee q) \equiv \neg p \wedge \neg q$

$\leftrightarrow$

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Rules of Inference:- Proofs in mathematics are valid arguments that establish the truth of mathematical statements.

Argument:- An argument in propositional logic is a sequence of propositions. (or) A sequence of statements that end with a conclusion. All but the final proposition in the argument are called premises and the final proposition is called the conclusion.

An argument is valid if the truth of all its premises implies that the conclusion is true.

Valid argument:- A valid argument is a sequence of propositions $P_1, P_2, \ldots, P_n$ called premises, together with a proposition C called the conclusion, such that the implication $P_1 \wedge P_2 \wedge \ldots \wedge P_n \Rightarrow C$ is a tautology.

An argument is valid if, whenever each of its premises $P_1, P_2, \ldots P_n$ is true, its conclusion C is also true.

Each inference can be written as an implication, as

(Conjunction of premises) $\longrightarrow$ Conclusion

The conclusion is a valid one, if the implication is a tautology.

P_1
P_2
$\vdots$
P_n
} sequence of premises

$\therefore C \rightarrow$ Conclusion

An argument form in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is valid if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

Rules of inference provide the justification of the steps used in a proof.

[inference - is an idea or conclusion that is drawn from evidence and reasoning.]

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Table 1. Rules of Inference for propositional logic

	Rule of Inference	Tautology	Name
①	$\begin{array}{l} p \\ p \to q \\ \hline \therefore q \end{array}$	$[p \wedge (p \to q)] \to q$	Modus ponens (or) Law of detachment
②	$\begin{array}{l} \neg q \\ p \to q \\ \hline \therefore \neg p \end{array}$	$[\neg q \wedge (p \to q)] \to \neg p$	Modus tollens (or) Law of contraposition
③	$\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore p \to r \end{array}$	$[(p \to q) \wedge (q \to r)] \to (p \to r)$	Hypothetical Syllogism (or) Transitive rule
④	$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$[(p \vee q) \wedge q] \to q$	Disjunctive Syllogism
⑤	$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \to (p \vee q)$	Addition
⑥	$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \to p$	Simplification
⑦	$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$[(p) \wedge (q)] \to (p \wedge q)$	Conjunction
⑧	$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$[(p \vee q) \wedge (\neg p \vee r)] \to (q \vee r)$	Resolution

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Rules of inference :-

Establish the validity of some relatively simple argument forms, called rules of inference. These rules of inference can be used as building blocks to construct more complicated valid argument forms.

Example: p

$p \to q$

$\therefore q$

Using this notation, the hypotheses are written in a column, followed by a horizontal bar, followed by line that begins with the therefore ($\therefore$) symbol and ends with a conclusion

Q. Check the validity of the following argument.

(a) It is below freezing now.
Therefore, it is either below freezing or raining now.

↓ Let p be the proposition "It is below freezing" and
q be the proposition "It is raining now". Then this argument is of the form.

p

$\therefore p \vee q$ This is the addition rule.

(b) It is below freezing and raining now.

↓ let p: below freezing
q: It is raining now. Then this argument is of the form,

$p \wedge q$

$\therefore p$ This argument uses the Simplification rule.

(c) If John has a B in calculus, he will graduate
John has a B in Calculus.
Therefore, he will graduate.

↓ let p: John has a B in Calculus
q: He will graduate
Then this argument is of the form

$p \to q$

p

$\therefore q$ Modus Ponens

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(d) If Harry is dentist, then Harry drills teeth.
Harry does not drill teeth.
Therefore, Harry is not a dentist.

Let p: Harry is dentist
q: Harry drills teeth
then this argument is of the form

$$\begin{array}{l} p \to q \\ \neg q \\ \hline \therefore \neg p \end{array}$$ Modus tollens

(e) Either elephants are black or monkeys are green
Elephants are grey.
Therefore, monkeys are green.

Let p: Elephants are black
q: Monkeys are green
then this argument is of the form

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$ Disjunctive syllogism

(f) If Mary is a senior, then Mary wears a pin
If Mary wears a pin, then Mary will graduate.
Therefore, If Mary is a senior, then Mary will graduate.

Let p: Mary is a senior
q: Mary wears a pin
r: Mary will graduate
Then this argument is of the form

$$\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore p \to r \end{array}$$ Hypothetical syllogism

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Q. Check the validity of the following argument.

If Ram has completed B.E computer science or M.B.A, then he is assured of a good Job.

If Ram is assured of a good Job, he is happy.

Ram is not happy.

Therefore, Ram is not completed M.B.A.

Let p: Ram has completed B.E computer science

q: Ram has completed M.B.A.

r: Ram is assured of a good Job.

s: Ram is happy.

The given premises are: 1. $(p \vee q) \rightarrow r$

2. $r \rightarrow s$

3. $\neg s$

1. $(p \vee q) \rightarrow r$	premise 1
2. $r \rightarrow s$	premise 2
3. $(p \vee q) \rightarrow s$	Hypothetical Syllogism using (1) & (2)
4. $\neg s$	premise 3
5. $\neg(p \vee q)$	Modus tollens using (3) & (4)
6. $\neg p \wedge \neg q$	Using De Morgans Law
7. $\neg q$	Simplification using (6).

Therefore, the conclusion is $\neg q$.

Fallacies:- Several common fallacies arise in incorrect arguments. These fallacies resemble rules of inference but are based on contingencies rather than tautologies.

Example: The proposition $[(p \rightarrow q) \wedge q] \rightarrow p$ is not a tautology. because it is false when p is false and q is true.

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Predicates and Quantifiers:-

Predicate Logic:- deals with predicates, which are propositions containing variables. It is an extension of propositional logic. It adds the concept of predicates and quantifiers to better capture the meaning of statements that cannot be adequately expressed by propositional logic. (meaning satisfactory or acceptable extent.)

Example: "Every person who is 18 years or older, is eligible to vote."
The above statement cannot be adequately expressed using only propositional logic. Therefore we need a more powerful type of logic.

Q. What is a predicate?

Consider the statement, "x is greater than 3" has two parts.
The first part, the variable x, is the subject of the statement.
The second part, "is greater than 3", is the predicate.
It refers to the property that the subject of the statement can have.

The statement "x is greater than 3" can be denoted by $P(x)$ where P denotes the predicate "is greater than" and x is the variable.
The predicate P can be considered as a function. It tells the truth value of the statement $P(x)$ at x. Once a value has been assigned to the variable x, the statement $P(x)$ becomes a proposition and has a truth or false value.

In general, a statement involving n variables $x_1, x_2, \ldots, x_n$ can be denoted by $P(x_1, x_2, \ldots, x_n)$. Here P is also referred to as n-place predicate or n-ary predicate.

Example (1): Let $P(x)$ denote the statement "$x > 10$". What are the truth values of $P(11)$ and $P(5)$?

Sol: $P(11)$ is equivalent to the statement $11 > 10$, which is True.
$P(5)$ is equivalent to the statement $5 > 10$, which is False.

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Example ②:- Let $R(x,y)$ denote the statement "$x = y+1$". What is the truth of the propositions $R(1,3)$ and $R(2,1)$?

Sol: $R(1,3)$ is the statement $1 = 3+1$, which is False.

$R(2,1)$ is the statement $2 = 1+1$, which is True.

We shall symbolize/represent a predicate by capital letters and subject by small letter.

① "Florida is a State" can be represented as $S(f)$.

② "Joe is a mathematician" represented as $M(j)$.

n-place predicate:

Examples: ① Ram is a student – $S(r)$ – 1-place predicate

② Jack is taller than Jill – $T(j_1, j_2)$ – 2-place predicate.

③ Sushan sits between Jack and Jill – $S(s_1, j_1, j_2)$ – 3 place predicate.

predicate: A predicate is an expression of one or more variables defined on some specific domain. A predicate with variables can be made a proposition by either assigning a value to the variable or by quantifying the variable.

Examples: ① $R(x)$: x is a rational number

② $Q(y)$: $y > 5$

③ $S(x,y)$: $x + y = 5$

④ $L(x)$: x is a lawyer

⑤ $C(y)$: y is a computer programmer

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Quantifiers :-

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value. However, there is another important way, called quantification, to create a proposition from a proposition function.

Quantification expresses the extent to which a predicate is true over a range of elements. In English, all, some, many, none, and few are used in quantifications.

There are two types of quantifiers. They are :

① Universal quantifier : Which tells us that a predicate is true for every element under consideration.

② Existential quantifier :- Which tells us that there is one or more element under consideration for which the predicate is true.

① Universal quantifier :-

Many mathematical statements assert that a property is true for all values of a variable in a perticular domain, called the domain of discourse (or the universe of discourse) or domain.

Def: The universal quantification of $P(x)$ is the statement "$P(x)$ is true for all values of x in the domain".

The notation $\forall x\, P(x)$ denotes the universal quantification of $P(x)$.
Here $\forall$ is called the universal quantifier.

We read $\forall x\, P(x)$ as "For all x $P(x)$" or "For every x $P(x)$" or "For each x $P(x)$".

$\forall x$ quantifies a conditional statement.
An element for which $P(x)$ is false is called a counterexample of $\forall x\, P(x)$.

$\forall x$ may be expressed as: All, for each, given any, for any, for arbitrary.

NOTE: It is best to avoid using 'for any'. It is ambiguous as to whether "any" means every or some.

Examples: ① All roses are red.
This can be understood as ~~for~~ every x.
If x is rose then x is red.
$R(x)$: x is rose
$P(x)$: x is red

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The statement as $[R(x) \rightarrow P(x)]$ before use $\forall x$

$\therefore \quad \forall x\,[R(x) \rightarrow P(x)]$.

② Every apple is red

For every x, if x is apple then x is red.

$A(x)$: x is apple

$R(x)$: x is red.

$\forall x\,[A(x) \rightarrow R(x)]$.

③ Any integer is positive or negative

For any x if x is an integer then x is either positive or negative.

$I(x)$: x is an integer

$P(x)$: x is positive or negative.

$\forall x\,[I(x) \rightarrow P(x)]$.

The symbols (x) or $(\forall x)$ are called universal quantifiers.

② Existential quantifier :-

Many mathematical statements assert that there is an element with a certain property.

A proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

Definition: The existential quantification of $P(x)$ is the proposition "There exists an element x in the domain such that $P(x)$".

We use the notation $\exists x\,P(x)$ for the existential quantification.

Here $\exists$ is called existential quantifier.

Without specifying the domain, the stmt $\exists x\,P(x)$ has no meaning.

$\exists x\,P(x)$ is read as ① There exists an x such that $P(x)$

② There is an x such that $P(x)$

③ There is atleast one x such that $P(x)$

④ for some x $P(x)$.

$\exists x$ quantifies a conjunction

Existential quantification may be expressed as:
There exists, for some, for at least one, or there is.

Examples: ① There exists a man

i: There exists an x such that x is a man.

$M(x)$: x is a man

$\exists x\ M(x)$

② Some men are clever.

There exists an x such that x is a man and x is clever.

$M_1(x)$: x is a man

$M_2(x)$: x is a clever.

$\exists x\ [M_1(x) \wedge M_2(x)]$.

Sentence	meaning	
① $\forall x\ P(x)$	All true	1
② $\exists x\ P(x)$	At least one true	5
③ $\neg[\exists x\ P(x)]$	None true	7
④ $\forall x[\neg P(x)]$	All False	3
⑤ $\exists x\ [\neg P(x)]$	Atleast one False	6
⑥ $\neg[\exists x[\neg P(x)]]$	None False	8
⑦ $\neg[\forall x\ P(x)]$	Not all true	2
⑧ $\neg[\forall x[\neg P(x)]]$	Not all False.	4

From the above table, We can conclude that

"All true" means "None False"

"All False" means "None True"

"Not all true" means "At least one False"

"Not all False" means "At least one True".

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The above eight expressions can be grouped as: (Equivalent)

"All true" $\{\forall x\, P(x)\} \equiv \{\neg\{\exists x\, \{\neg P(x)\}\}\}$ = "None False"

"All False" $\{\forall x \{\neg P(x)\}\} \equiv \{\neg\{\exists x\, P(x)\}\}$ = "None True".

"Not all true" $\{\neg\{\forall x\, P(x)\}\} \equiv \{\exists x [\neg P(x)]\}$ = "At least one False".

"Not all False" $\{\neg\{\forall x \{\neg P(x)\}\}\} \equiv \{\exists x\, P(x)\}$ = "At least one True".

Quantifiers:

	Statement	When true?	When False
①	$\forall x\, P(x)$	$P(x)$ is true for every x	There is an x for which $P(x)$ is false.
②	$\exists x\, P(x)$	There is an x for which $P(x)$ is true	$P(x)$ is false for every x.

Negating Quantified Expressions:-

	Statement	Negation
① All true	$\forall x\, P(x)$	$\exists x [\neg P(x)]$ At least one False
② At least one False	$\exists x [\neg P(x)]$	$\forall x\, P(x)$ All true
③ All False	$\forall x [\neg P(x)]$	$\exists x\, P(x)$ At least one true
④ At least one true	$\exists x\, P(x)$	$\forall x [\neg P(x)]$ All False.

Predicate Calculus:-

The area of logic that deals with predicates and quantifiers is called the predicate calculus.

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Q. Write the following statements in Symbolic form:

(a) Some thing is good
(b) Every thing is good
(c) Nothing is good
(d) something is not good.

Let $G(x)$ be a good.

(a) $\exists x\,(G(x))$
(b) $\forall x\,(G(x))$
(c) $\forall x\,(\neg(G(x)))$
(d) $\exists x\,(\neg(G(x)))$.

Q. Write each of the following in Symbolic form:

(a) All men are good $\quad \forall x\,[M(x) \rightarrow G(x)]$

(b) NO men are good $\quad \neg\exists x\,[M(x) \wedge G(x)] \equiv \forall x\,(M(x) \rightarrow \neg G(x)) \equiv \neg\exists x\,[M(x) \wedge G(x)]$

(c) Some men are good $\quad \exists x\,[M(x) \wedge G(x)]$

(d) Some men are not good $\quad \exists x\,[M(x) \wedge \neg G(x)]$.

$M(x)$: x is a man
$G(x)$: x is good.

---

predicate logic uses the following need features:

Variables: x, y, z
predicates: $P(x), Q(x)$
Quantifier — Universal — $\forall x\,P(x)$
Existential — $\exists x\,Q(x)$

propositional functions are a generalization of propositions.
- They contain variables and a predicate e.g. $P(x)$.
- Variables can be replaced by elements from their domain

Ex: Let $P(x)$: $x > 0$ domain better integers.
$P(-3)$ is false
$P(0)$ is false
$P(3)$ is True.

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Q. Translate each of the following statements into symbols, using quantifiers, variables, and predicate symbols.

(a) All birds can fly

Let $B(x)$: x is a bird

$F(x)$: x can fly

Then the stmt can be written as $\forall x [B(x) \rightarrow F(x)]$

where universe of discourse is the set of all birds.

(b) Some babies are illogical.

Let $B(x)$: x is a baby

$I(x)$: x is illogical

The given stmt can be symbolized as $\exists x [B(x) \wedge I(x)]$

The universe of discourse is the set of all babies.

(f) All babies are illogical.

$\forall x [B(x) \rightarrow I(x)]$

(i) Some men are not gaints

$\exists x [M(x) \wedge (\neg G(x))]$

(c) Some men are gaints

Let $M(x)$: x is a man

$G(x)$: x is a gaint

Stmt can be written as $\exists x [M(x) \wedge G(x)]$

where universe of discourse is the set of men in the world.

(g) All men are gaints

$\forall x [M(x) \rightarrow G(x)]$

(h) No men are gaints

$\forall x [(M(x)) \rightarrow \neg G(x)]$

(d) There is a student who likes mathematics but not history.

Let $S(x)$: x is a student

$M(x)$: x likes mathematics

$H(x)$: x likes history

The stmt can be written as $\exists x [S(x) \wedge M(x) \wedge H(x)]$

where universe of discourse is the set of all students at your college.

(e) Not all birds can fly.

$\neg(\forall x)[B(x) \rightarrow F(x)]$ or $\exists x [B(x) \wedge (\neg F(x)]$

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(J) If x is a man, then x is gaint.

Let $M(x)$: x is a man

$G(x)$: x is gaint

Symbolized as $M(x) \rightarrow G(x)$.

(K) x is an odd integer and x is prime.

$I(x)$: x is an odd integer

$P(x)$: x is prime

The Stmt can be symbolized as $I(x) \wedge P(x)$.

(l) For all integers x, x is odd and x is prime.

Let $I(x)$: x is odd

$P(x)$: x is prime

The Stmt can be written as $\forall x [I(x) \wedge P(x)]$

(m) There is an integer x_0 such that x is odd and x is prime.

$\exists x [I(x) \wedge P(x)]$

(n) Not every actor is talented who is famous.

Let $A(x)$: x is an actor

$T(x)$: x is talented

$F(x)$: x is famous

$\exists x [A(x) \wedge F(x) \wedge (\neg T(x))]$

At least one actor who is famous is not talented.

(o) x is rational implies that x is real.

Let $R_1(x)$: x is rational

$R_2(x)$: x is real

$R_1(x) \rightarrow R_2(x)$.

(p) Not every graph is planar

Let $G(x)$: x is a graph

$P(x)$: x is planar

The Stmt can be written as $\neg [\forall x (G(x) \rightarrow P(x))]$

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(q) Some numbers are rational.

Let $N(x)$: x is a number

$R(x)$: x is rational

The stmt can be written as $\exists x [N(x) \wedge R(x)]$

(r) Some numbers are not rational

$\exists x [N(x) \wedge (\neg R(x))]$

(s) Not all numbers are rational.

$\neg [\forall x [N(x) \rightarrow R(x)]]$

(t) If some students are lazy, then all students are lazy.

$S(x)$: x is a student

$L(x)$: x is lazy.

some students are lazy

$[\exists x [S(x) \wedge L(x)]]$

All students are lazy

$\forall x [S(x) \rightarrow L(x)]$

∴ The final stmt can be written as

$(\exists x [S(x) \wedge L(x)]) \rightarrow (\forall x [S(x) \rightarrow L(x)])$.

---

(1) Examples:

$P(x)$ — x is free

$P(5)$ — x is bound to 5

$\forall x P(x)$ — x is bound by quantifier

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Other quantifiers:-

(1) Uniqueness quantifier:-

Definition: "There exists a unique x such that $P(x)$ is true".
denoted by $\exists!$ or $\exists_1$. $\exists! x\, P(x)$ or $\exists_1 x\, P(x)$.

other phrases for uniqueness quantification include "There is exactly one" and "There is one and only one".

Precedence of Quantifiers:-

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus.

For example, $\forall x\, P(x) \vee Q(x)$ is the disjunction of $\forall x\, P(x)$ and $Q(x)$.
$\hookrightarrow \equiv (\forall x\, P(x)) \vee Q(x)$.

Binding Variables:-

When a quantifier is used on the variable x, we say that this occurance of the variable is **bound**. Example: $P(5)$ – x is bound to 5

(1) An occurance of a variable that is not bound by a quantifier (or) set equal to a particular value is said to be **free**. Ex: $P(x)$ – x is free

All the variables that occur in a propositional function must be bound or set equal to a particular value to turn it onto a proposition. This can be done using a combination of universal quantifiers, existential quantifiers and value assignments.

Example: $\exists x\,(x+y=1)$, Ex: $\forall x\, P(x)$ – x is bound by quantifier.
existential quantifier $\exists x$, but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable. $\therefore$ $\exists x\,(x+y=1)$, x is bound, but y is free.

The part of a logical expression to which a quantifier is applied is called the **scope** of this quantifier.

A variable is free if it is outside the scope of all quantifiers.

$\exists x\,(P(x) \wedge Q(x)) \vee \forall x\, R(x)$, all variables are bound. The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$. because $\exists x$ is applied only to $P(x) \wedge Q(x)$. and not to the rest of the stmt.

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i.e the existential quantifier binds the variable x in $P(x) \wedge Q(x)$. and the universal quantifier $\forall x$ binds the variable x in $R(x)$.

observe that using two different variables x and y, as $\exists x(P(x) \wedge Q(x)) \vee \forall y R(y)$. because the scopes of the two quantifiers do not overlap.

Logical equivalences involving quantifiers:-

Definition:- stmts involving predicates and quantifiers are logically equivalent if and only if they have the same truth value no matter which predicates are substituted into these stmts and which domain of discourse is used for the variables in these propositional functions.

$S \equiv T$, indicates that S and T involving predicates and quantifiers are logically equivalent.

Example: $\forall x(P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$.

↳ true — also — true

Translating from English into logical expressions:-

Q. Express the stmt " Every student in this class has studied calculus" Using predicates and quantifiers.

A We rewrite the stmt as:

"For every student in this class, that student has studied Calculus".

Next, We introduce a variable x, so the above stmt becomes

For every student x in this class, x has studied calculus.

Let $C(x)$: x has studied Calculus.

If the domain for x consists of the students in the class, so, we can translate the stmt as $\forall x\, C(x)$.

Q. For every person x, if x is a student, then x has visited Mexico or x has visited Canada.

Let $M(x)$: x has visited Mexico

$C(x)$: x has visited Canada

$S(x)$: x is a student

$$\forall x[S(x) \rightarrow [M(x) \vee C(x)]]$$

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Using quantifiers in System specifications :-

Q. Use predicates and quantifiers to express the system specifications

① " Every mail message larger than one megabyte will be compressed".

② " If a user is active, at least one network link will be available".

✱

① Let $S(m, y)$ be Mail message m is larger than y megabytes.
Where the variable x has the domain of all mail messages and the variable y is a positive real number.
$C(m)$ denote Mail message m will be compressed.
So, the stmt can be written as $\forall m[S(m,1) \rightarrow C(m)]$

② Let $A(u)$: User u is active
Where u has the domain of all users.
$S(n,x)$: Network link n is in state x
Where n has the domain of all network links and x has the domain of all possible states for a network link.
Then the specification " If a user is active, at least one network link will be available".
Can be represented as $\exists u\, A(u) \rightarrow \exists n\, S(n, \text{available})$.

problem solving :-

Q①:- All Lions are fierce.
Some Lions do not drink coffee. } premises

Conclusion: Some fierce creatures do not drik coffee.

Let $P(x)$: x is a lion
$Q(x)$: x is fierce
$R(x)$: x drinks coffee
Assuming that the domain consists of all creatures.
We express the above stmts as

$\forall x\,[P(x) \rightarrow Q(x)]$
$\exists x[P(x) \wedge \neg R(x)]$
$\exists x[Q(x) \wedge \neg R(x)]$

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~~Nested Quantifiers~~:-

Rules of Inference for Quantified statements:-

	Rule of Inference	Name
①	$\forall x\, P(x)$ $\therefore P(c)$	Universal instantiation
②	$P(c)$ for any arbitrary c $\therefore \forall x\, P(x)$	Universal generalization
③	$\exists x\, P(x)$ $\therefore P(c)$ for some element c	Existential instantiation
④	$P(c)$ for some element c $\therefore \exists x\, P(x)$	Existential ~~instantiation~~ generalization.

---

① Universal instantiation / specification:-

The rule of inference used to conclude that $P(c)$ is true, where c is a particular member of the domain, given the premise $\forall x\, P(x)$.

$$\frac{\forall x\, P(x)}{\therefore P(c)}$$

Example ①:- Universe is the set of humans.

Let $M(x)$: x is mortal, then if $\forall x\, M(x)$ is true. i.e All men are mortal, they as per this rule we can conclude that "Socrates is mortal".

② All Women are wise

so Lisa is wise. where Lisa is a member of the domain of all women.

③ Universal generalization:-

$\forall x P(x)$ is true, given the premise that $P(c)$ is true for all elements c in the domain.

This rule holds - provided we know $P(c)$ is true for each element in the Universe.

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x\, P(x)}$$

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③ Existential instantiation:-

There is an element c in the domain for which P(c) is true if we know that $\exists x\, P(x)$ is true.

We cannot select an arbitrary value of c.

$$\frac{\exists x\, P(x)}{\therefore P(c) \text{ for some element } c}$$

④ Existential generalization:-

$\exists x\, P(x)$ is true when a particular element c with P(c) true is known.

$$\frac{P(c) \text{ for some element } c}{\therefore \exists x\, P(x).}$$

Problems:-

① Consider the argument

All men are fallible

All kings are men

Therefore, all kings are fallible.

Let M(x): x is a man

K(x): x is a king

F(x): x is fallible

Then the above argument is symbolized as

1. $\forall x\,[M(x) \rightarrow F(x)]$
2. $\forall x\,[K(x) \rightarrow M(x)]$

3. $\therefore \forall x\,[K(x) \rightarrow F(x)]$

A formal proof is as follows:

	Assertion	Reason
1.	$\forall x\,[M(x) \rightarrow F(x)]$	premise 1
2.	$M(c) \rightarrow F(c)$	Step 1 and Universal Specification/instantiation
3.	$\forall x\,[K(x) \rightarrow M(x)]$	premise 2
4.	$K(c) \rightarrow M(c)$	Step 3 and Universal instantiation
5.	$K(c) \rightarrow F(c)$	steps 2 & 4 & ~~Simplification~~ transitive rule (HS).
6.	$\forall x\,[K(x) \rightarrow F(x)]$	Step 5 and hypothetical syllogism.

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(2) Lions are dangerous animals
There are Lions.
Therefore, there are dangerous animals.

L(x) : x is a Lion
D(x) : x is dangerous

1 $\forall x [L(x) \rightarrow D(x)]$
2 $\exists x\, L(x)$
---
3 $\therefore \exists x\, D(x)$.

Formal proof:-

	Assertion	Reasons
1.	$\exists x\, L(x)$	premise 2
2.	$L(a)$	Step 1 & Existential instantiation
3.	$\forall x [L(x) \rightarrow D(x)]$	~~Step 3 and universal instantiation~~ premise 1
4.	$L(a) \rightarrow D(a)$	Step 3 & universal instantiation
5.	$D(a)$	Step 2 & 4 & Hypothetical syllogism
6.	$\exists x\, D(x)$	Step 5 & Existential generalization.

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Nested quantifiers :-

Two quantifiers are nested if one is within the scope of the other.

Example: ① $\forall x \exists y \underbrace{(x+y=0)}_{P(x,y)}$

$\underbrace{\qquad\qquad}_{Q(x)}$

Everything within the scope of a quantifier. is the propositional function.

$\forall x \, Q(x)$

$\hookrightarrow \exists y \, P(x,y)$

$P(x,y)$ is $(x+y=0)$

Nested quantifiers commonly occur in mathematics and computer science.

② $\forall x \forall y (x+y = y+x)$ x and y are variables in the domain of all real numbers.

For all real numbers x and y, $x+y = y+x$

similarly, $\forall x \forall y \forall z (x+(y+z) = (x+y)+z)$ ⇒ Associative law for addition of real numbers.

③ $\forall x \forall y ((x>0) \wedge (y<0) \rightarrow (xy<0))$

Where the domain for both variables consists of all real numbers.

For every real numbers x and y, if x is positive and y is negative then xy is negative.

The product of a positive real number and a negative real number is always a negative real number.

Note: In working with quantifications of more than one variables it is sometimes helpful to think in terms of nested loops.

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The order of Quantifiers :-

Many mathematical statements involve multiple quantifications of propositional functions involving more than one variable.

Example ①: Let $P(x,y)$: $x+y=y+x$. What are the truth values of the quantifications $\forall x \forall y\, P(x,y)$ and $\forall y \forall x\, P(x,y)$. Where the domain for all variables consists of all real numbers?

Ⓐ $\forall x \forall y\, P(x,y)$ denotes the proposition

"For all real numbers x, for all real numbers y, $x+y=y+x$".

because $P(x,y)$ is true for all real numbers x and y, the proposition $\forall x \forall y\, P(x,y)$ is true.

Theorem 1 The stmt says " The order of nested Universal quantifiers in a stmt without other quantifiers can be changed without changing the meaning of the quantified stmt.

Theorem 2: The order of nested existential quantifiers in a stmt without other quantifiers can be changed without changing the meaning of the quantified stmt.]

Table: Quantification of two variables

Stmt	When True?	When False?
① $\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x,y for which $P(x,y)$ is false.
② $\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
③ $\exists x \forall y\, P(x,y)$	There is an x for which $P(x,y)$ is true for every y	For every x there is a y for which $P(x,y)$ is false.
④ $\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x,y for which $P(x,y)$ is true.	$P(x,y)$ is false for every pair x,y.

Ex: Let $Q(x,y,z)$: $x+y=z$. What are the truth values of the stmts $\forall x \forall y \exists z\, Q(x,y,z)$ and $\exists z \forall x \forall y\, Q(x,y,z)$, where domain of all variables consists of all real numbers?

Ⓐ Let x & y are assigned variables. $\forall x \forall y \exists z\, Q(x,y,z)$ means " For all real numbers x and for all real numbers y, there is a real number z such that $x+y=z$".

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Translating Mathematical Stmts into stmts involving Nested quantifiers:-

Q① Translate the stmt "The sum of two positive integers is always positive" into a logical expression.

Sol "For every two integers, if these integers are both positive, then the sum of these integers is positive".

We introduce the variables x and y. So we can state the stmt as "For all positive integers x and y, $x+y$ is positive".

$\Rightarrow \forall x \forall y ((x>0) \wedge (y>0) \rightarrow (x+y>0))$

domain: x and y are variables consists of all integers.

Then the stmt: The sum of two positive integers is always positive". becomes "For every two positive integers, the sum of these integers is positive".

We can express this as $\forall x \forall y (x+y>0)$,

Where the domain for both variables consists of all positive integers.

② "Every real number except zero has a multiplicative inverse"

(A multiplicative inverse of a real number x is a real number y such that $xy=1$).

Sol "For every real number x except zero, x has a multiplicative inverse".

We can rewrite this as "For every real number x, if $x \neq 0$, then there exists a real number y such that $xy=1$".

This can be rewritten as $\forall x (x \neq 0) \rightarrow \exists y (xy=1)$.

③ "If a person is female and is a parent, then this person is someone's mother" with a domain consisting of all people.

Sol "For every person x, if person x is female and person x is a parent, then there exists a person y such that person x is the mother of person y"

Let $F(x)$: x is female

$P(x)$: x is a parent

$M(x,y)$: x is the mother of y

The stmt can be written as $\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x,y))$

The final stmt is $\forall x \exists y [(F(x) \wedge P(x)) \rightarrow M(x,y)]$

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Negating Nested Quantifiers:-

Statements involving nested quantifiers can be negated by successively applying the rules for negating statements involving a single quantifier.

Example ①: Express the negation of the stmt $\forall x \exists y (xy = 1)$.

Apply De. Morgan's Laws for quantifiers.

$$\neg(\forall x \exists y (xy = 1)) = \exists x \forall y (xy \neq 1)$$

Quantifiers as Conjunctions and Disjunctions:-

$$\forall x P(x) = P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) = P(1) \vee P(2) \vee P(3).$$

If U consists of the integers 1, 2 and 3.

Counter example:-

An element for which $P(x)$ is false is called a counterexample of $\forall x P(x)$.

Example ②: Let $Q(x) : x < 2$. What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

$Q(x)$ is not true for every real number x, because, for instance, $Q(3)$ is false. i.e $x = 3$ is a counterexample for the stmt $\forall x Q(x)$. Thus $\forall x Q(x)$ is false.

Q. Translate the following stmt into English.

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

Where $C(x)$: x has a computer

$F(x, y)$: x and y are friends

Domain of x and y: all students.

For every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends.

In other words, Every student has a computer or has a friend that has a computer.

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Q. Translating the following statements into logical expression.

(a) Everyone has exactly one best friend.

$B(x,y)$: y is the best friend of x.

∴ For every person x, person x has exactly one best friend.

Introducing the Universal quantifier, then the statement as

$\forall x$ (person x has exactly one best friend)

Where the domain consists of all people.

We rewrite the statement as

For all x, there is y who is the best friend of x and for every person z, if person z is not person y, then z is not the best friend of x.

$$\forall x \exists y (B(x,y) \wedge \forall z [(z \neq y) \rightarrow \neg B(x,z)])$$

Domain of x, y, and z: all people.

(b) There is a woman who has taken a flight on every airline in the world.

∴ Let $T(w,f)$: w has taken flight

$A(f,a)$: flight is on airline

$$\exists w \forall a \exists f (T(w,f) \wedge A(f,a))$$

Domain of w: all people

Domain of f: all flights

Domain of a: all airlines.

[In other way, $\exists w \forall a \exists f \, R(w,f,a)$

where $R(w,f,a)$ is "w has taken f on a"]

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Introduction to proofs:-

① A theorem (Can also be referred to as facts or results.) is a statement that can be shown to be true.

A proof is a valid argument that establishes the truth of a mathematical statement. A proof can use the hypothesis of the theorem.

A theorem is true with a proof (valid argument) using:
- Definitions
- previously proven theorems or other theorems.
- Rules of Inference
- Axioms — A stmt that is assumed to be true.

Applications:- ① Verifying that Computer programs are correct
② Establishing that operating systems are secure.
③ Making inferences in artificial Intelligence.
④ Showing that system specifications are consistent.

- A Lemma is a 'helping theorem' or a result which is needed to prove a theorem.
- A Corollary is a result which follows directly from a theorem.
- Less important theorems are sometimes called propositions.
- A Conjecture is a statement that is being proposed to be true. Once a proof of a conjecture is found, it becomes a theorem. It may turn out to be false. so they are not theorems.

Understanding how Theorems are Stated./ Forms of Theorems

- Many theorems assert that a property holds for all elements in a domain, such as the integers, the real numbers.
- often the universal quantifier (needed for a precise stmt of a theorem) is omitted by standard mathematical convention.

Example: If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$. means that

For all positive real numbers x and y,
if $x > y$, then $x^2 > y^2$.

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Methods of proving theorems:-

(1) Direct proof:-

In a direct proof, we show that conditional statement $p \rightarrow q$ is true. We assume that p is true and show that q must also be true, so that the combination p is true and q false never occcers.

In a direct proof, we assume that p is true and use axioms, definitions, and previously proven theorems, together with rules of inference, finally to show that q must also be true.

→ Sequence of steps leading from the hypothesis to the conclusion. [Direct proof lead from hypothesis of a theorem to the conclusion].

Definition (1): The integer n is even if there exists an integer k such that $n = 2k$, and n is odd if there exists an integer k such that $n = 2k+1$.
(Every integer is either even or odd and no integer is both even and odd).

Example (1): Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd".

Sol: Assume n is an odd integer

Then $n = 2k+1$ for some integer k

We can square both sides of the equation $n = 2k+1$

$$n^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$$
$$= 2m+1$$
where $m = 2k^2+2k$

Hence n^2 is odd.

∴ We have proved that if n is an odd integer, then n^2 is odd.

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Example ②: Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square.

Sol: Let r and s are two integers,

then $m = r^2$ and $n = s^2$

now $nm = s^2 r^2$

$= (sr)^2$

$= (t)^2$ where $t = sr$

Hence proved.

∴ if m and n are both perfect squares, then nm is also a perfect square.

② proof by Contraposition :-

proof by contraposition is a type of indirect proof.

We know that $p \to q \equiv \neg q \to \neg p$.

This means that the conditional statement $p \to q$ can be proved by showing that its contrapositive $\neg q \to \neg p$ is true.

In a proof by contraposition of $p \to q$, we assume that $\neg q$ is true (or we take $\neg q$ as a hypothesis) and using axioms, definitions, and previously proven theorems, together with rules of inference, and we show that $\neg p$ is true.

Example ①:- prove that if n is an integer and $3n+2$ is odd, then n is odd. (p / $\neg q$)

Sol: Assume that n is even (negation). So, n can be expressed as $2k$ for some integer k (definition of even)

$n = 2k$

∴ $3n + 2 = 3(2k) + 2$

$= 6k + 2$

$= 2(3k+1)$

So, $3n+2$ is even ($\neg p$)

Hence proved.

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Example ②:- prove that if $n = ab$, where a and b are positive integers, then $a \le \sqrt{n}$ or $b \le \sqrt{n}$.

Sol: $n = ab$ then $a \le \sqrt{n}$ or $b \le \sqrt{n}$

Now assume that the above statement is false

$\neg(a \le \sqrt{n} \text{ or } b \le \sqrt{n})$

$\equiv$

$a > \sqrt{n}$ and $b > \sqrt{n}$

$ab > \sqrt{n} \cdot \sqrt{n}$

$ab > n$

ab is not equal to n i.e $ab \neq n$

which contradicts the statement $n = ab$.

③ proof by Contradiction:-(AKA reductio ad absurdum)

It is another type of indirect proof. $p \to q$

[To prove P, assume $\neg P$ and derive a contradiction such as $P \wedge \neg P$. Since we have shown that $\neg P \to F$ is true, it follows that the contrapositive $T \to P$ as holds.]

Assume P and $\neg Q$ are true. showing that q must also be true. This implies that q and $\neg q$ are both true. This is a contradiction. (or) Assume P and $\neg q$ is true. Showing that $\neg P$ must also be true. This implies that $P \& \neg P$ are both true. This is a contradiction.

Example ①: prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Definition: The real number r is rational if there exist integers p and q with $q \neq 0$ such that $r = p/q$. A real number that is not rational is called irrational.

Let p be the proposition "$\sqrt{2}$ is irrational".

$\neg P$: $\sqrt{2}$ is rational, and thus $2 = \frac{a}{b}$ where a & b have no common factors. Thus $2 = \frac{a^2}{b^2} \Rightarrow 2b^2 = a^2$

and thus a^2 is even.

a^2 is even and so a is even.

let $a = 2c$ for some integer c

$2b^2 = 4c^2$ and $b^2 = 2c^2$, and b^2 is even, b must also even.

This is contradiction. our assumption must be false.

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Example ②: If $3n+2$ is odd, then n is odd. Give proof by contradiction.

Let p: $3n+2$ is odd & q: n is odd

TO construct a proof by contradiction, Assume $p \wedge \neg q$ are true.

i.e $3n+2$ is odd and n is even.

If n is even, there is an integer k such that $n = 2k$.

This implies that $3n+2 = 3(2k)+2 = 6k+2 = 2(3k+1) = \text{even} = \neg p$

because an integer is even iff it is not odd. Because $p \wedge \neg p$ are true, we have a contradiction.

④ Vacuous and Trivial proofs:-

Vacous proof:- If we know p is false then $p \to q$ is true as well. [or $p \to q$ is true when p is false]

Example: ① ~~If it is raining then~~ If I am both rich and poor then $2+2=5$.

② Let $P(n)$: $\forall n$ If $n>1$, then $n^2 > n$, where the domain is all integers. sol: Vacuously, $P(0)$ is true since $0>1 \xrightarrow{F} 0>0$ is true

Trivial proof:- If we know q is true, then $p \to q$ is true as well.

[or $p \to q$ is true when q is true]

Example ①: If it is raining, then $1=1$.

② Let $P(n)$: If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$ where the domain is all integers.

sol: Trivially, $P(0)$ is true since: $a^0 \geq b^0$ regardless to the hypothesis

⑤ Counterexample:- A statement of the form $\forall x\, P(x)$ to be false.

Ex ① Show that the stmt "Every positive integer is the sum of the squares of two integers" is false.

sol: 3 cannot be written as the sum of the squares of two integers. This ~~fact~~ statement is ~~false~~ true.

Ex ②: Let $Q(x)$ be the stmt '$x < 2$'. $\forall x\, Q(x)$ is false.

sol $Q(x)$ is not true for every real number x, for instance, $Q(3)$ is false. i.e $x=3$ is a counterexample for the stmt $\forall x\, Q(x)$.

Therefore, $\forall x\, Q(x)$ is false.

⑥ proofs of Equivalence:-

To prove a theorem that is a biconditional statement, i.e a stmt of the form $p \leftrightarrow q$, we show that $p \to q$ and $q \to p$ are both true. The validity is based on the tautology.

$(p \leftrightarrow q) \leftrightarrow [(p \to q) \wedge (q \to p)]$

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Example ①: prove the theorem "If n is a positive integer, then n is odd if and only if n^2 is odd".

Sol: let p: n is odd and q: n^2 is odd

To prove this theorem, we need to show that $p \to q$ and $q \to p$ are true.

prove $p \to q$ is true: Assume n is an odd integer

Then $n = 2k+1$ for some integer k

square on both sides

$$n^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1 = 2m+1$$

where $m = 2k^2+2k$

$n^2 =$ odd

$\therefore$ $p \to q$ is true.

prove $q \to p$ is true:-

~~Assume n^2 is odd~~ ~~proof by contradiction example ①~~

~~Then $n = 2k+1$ for some integer k, n is odd.~~

~~$n^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k+1)$~~

~~$q \to p$ is also true~~

$\therefore$ ~~$p \to q$ is true and $q \to p$ is true~~.

Hence $p \leftrightarrow q$ is true.

The propositions $P_1, P_2, P_3, \ldots, P_n$ are equivalent. This can be written as $P_1 \leftrightarrow P_2 \leftrightarrow P_3 \cdots \leftrightarrow P_n$, which states that all n propositions have same truth values, and consequently, that for all i and j with $1 \le i \le n$ & $1 \le j \le n$, P_i and P_j are equivalent. one way to prove these mutually equivalent is to use the tautology.

$$[P_1 \leftrightarrow P_2 \leftrightarrow \cdots \leftrightarrow P_n] \leftrightarrow [(P_1 \to P_2) \wedge (P_2 \to P_3) \wedge \cdots \wedge (P_n \to P_1)].$$

if the conditional Stmts $P_1 \to P_2, P_2 \to P_3, \ldots P_n \to P_n$ can be shown to be true, then the propositions $P_1, P_2, \ldots P_n$ are all equivalent.

Example in Next Page. ①

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⑦ Mistakes in proofs:- There are many common errors made in constructing mathematical proofs.

Example①: What is Wrong with this proof that $1=2$?

Proof: We use these steps, where a and b are two equal positive integers.

Step	Reason
1. $a=b$	Given
2. $a^2=ab$	multiply both sides of (1) by a
3. $a^2-b^2=ab-b^2$	subtract b^2 from both sides of (2)
4. $(a-b)(a+b)=b(a-b)$	Factor both sides of (3)
5. $a+b=b$	Devide both sides of (4) by $a-b$
6. $2b=b$	Replace a by b in (5) because $a=b$ and simplify
7. $2=1$	Devide both sides of (6) by b

Sol: Every step is valid except for one, step 5.
There is an error in step 5.
$a-b=0$ by the premise and division by 0 is undefined.

Example②: What is Wrong with this proof?

Theorem:- If n^2 is positive, then n is positive.

proof: Suppose that n^2 is positive. Because the conditional stmt "If n is positive, then n^2 is positive" is true.
We can conclude that n is positive.

Sol: A counter example is supplied by $n=-1$ for which $n^2=1$ is positive, but n is negative.

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(1) ~~Logical~~ proofs of equivalence:-

Example (2): Show that these statements about the integer n are equivalent:

P_1 : n is even

P_2 : n−1 is odd

P_3 : n^2 is even.

Sol: We use a direct proof to show that $P_1 \to P_2$.

Suppose that n is even.

Then $n = 2k$ for some integer k

Consequently, $n-1 = 2k-1 = 2(k-1)+1$

$= 2m+1$ where $m = k-1$

$n-1$ = odd

We also use a direct proof to show that $P_2 \to P_3$:

Suppose n−1 is odd, then $n-1 = 2k+1$ for some integer k

Hence $n = 2k+2$

Square on both sides

$n^2 = (2k+2)^2 = 4k^2+8k+4 = 2(2k^2+4k+2)$

= even

TO prove $P_3 \to P_1$, we use a proof by contraposition.

We prove ~~Assume~~ that if n is not even, then n^2 is not even.

means that "if n is odd, then n^2 is odd. [Example problems in proof by contraposition].

This completes the proof.

$P_3 \to P_1 \equiv$ If n^2 is even, then n is even

$P + \neg P$

n is odd

$n = 2k+1$

$n^2 = (2k+1)^2 = 2m+1$ = odd. $\neg P$.

$\mapsto 2k^2+2k$

Hence proved.

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Proof methods and Strategy:-

1. Proof by cases.
2. Exhaustive proof.
3. Existence proofs
   - (a) Constructive
   - (b) Nonconstructive.
4. Uniqueness proofs.
5. proof Strategies
   - (a) Forward Reasoning
   - (b) Backward Reasoning
   - (c) Looking (Searching) for Counterexamples
   - (d) Adapting Existing proofs
6. The Role of open problems
7. Validity problem for Propositional and Predicate Logic.

(1) Proof by cases:-

Def:- A proof by cases must cover all possible cases that arise in a theorem.

- To prove a conditional statement of the form:

$$(P_1 \vee P_2 \vee \dots \vee P_n) \rightarrow q$$
$$\equiv \neg(P_1 \vee P_2 \vee \dots \vee P_n) \vee q$$
$$\equiv (\neg P_1 \wedge \neg P_2 \wedge \dots \wedge \neg P_n) \vee q$$
$$\equiv (\neg P_1 \vee q) \wedge (\neg P_2 \vee q) \wedge \dots \wedge (\neg P_n \vee q)$$
$$\equiv (P_1 \rightarrow q) \wedge (P_2 \rightarrow q) \wedge \dots \wedge (P_n \rightarrow q).$$

- Use the tautology

$$[(P_1 \vee P_2 \vee \dots \vee P_n) \rightarrow q] \leftrightarrow [(P_1 \rightarrow q) \wedge (P_2 \rightarrow q) \wedge \dots \wedge (P_n \rightarrow q)]$$

  can be used as a rule of inference.
- Each of the implications $P_i \rightarrow q$ is a case.
  where $i = 1, 2, \dots n$.
- prove a theorem by considering different Cases seperately.

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Example ①:- prove that if n is an integer, then $n^2 \geq n$.

Solution:

~~Proof~~: Consider 3 possible cases, when $n=0$, when $n \geq 1$ (positive), and when $n \leq -1$ (negative).

Case (i): If $n=0$, then $0^2=0$, we see that $0^2 \geq 0$. Thus, $n^2 \geq n$ is true in this case.

Case (ii): When $n \geq 1$, we multiply both sides by n, they $n \cdot n \geq n \cdot 1$. This implies that $n^2 \geq n$ for $n \geq 1$.

Case (iii): When $n \leq -1$. However, $n^2 \geq 0$, it follows that $n^2 \geq n$.

Because the inequality $n^2 \geq n$ holds in all three cases, we conclude that if n is an integer, then $n^2 \geq n$.

Example ②:- prove that $|xy| = |x||y|$, where x and y are real numbers.

[Definition: The absolute value of a, $|a|$, equals a when $a \geq 0$ and equals $-a$ when $a < 0$].

Sol: Break the theorem into some cases:

① x and y both nonnegative
② x nonnegative and y is negative
③ x negative and y is nonnegative
④ x and y both negative.

Check possible cases:

Case 1: We see that $P_1 \rightarrow q$ because $xy \geq 0$ when $x \geq 0$ and $y \geq 0$, so that $|xy| = xy = |x||y|$.

Case 2: To see that $P_2 \rightarrow q$, because $x \geq 0$ & $y < 0$, then $xy \leq 0$, so that $|xy| = -xy = x(-y) = |x||y|$.

Case 3: To see that $P_3 \rightarrow q$, because $x < 0$ & $y \geq 0$, then $xy \leq 0$, so that $|xy| = -xy = (-x)y = |x||y|$

Case 4: To see that $P_4 \rightarrow q$, because $x < 0$ & $y < 0$, then $xy \geq 0$. so that $|xy| = xy = (-x)(-y) = |x||y|$.

Because it is true for all four cases, so $|xy| = |x||y|$, where x and y are real numbers.

Leveraging proof by cases:- When it is not possible to consider all cases of a proof at the same time, a proof by cases should be considered. When should you use such a proof? We look for a proof by cases when there is no obvious way to begin a proof, but when extra information in each case helps move the proof forward.

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Without Loss of Generality (WLOG):-

How to shorten the proof by cases.

- If the same argument is used in different cases.
  - proved the cases together, without loss of generality (WLOG).
- Incorrect use of this principle can lead to errors.

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Common errors with Exhaustive proof and proof by Cases:-

A Common error of reasoning is to draw incorrect Conclusions from examples. No matter how many seperate examples are considered, a theorem is not proved by considering examples unless every posible case is covered.

The problem of proving a theorem is analogous to showing that a Computer program always produces the desired output. No matter how many input values are tested, unless all input values are tested, we cannot conclude that the program always produces the correct output.

Example 1: What is wrong with this 'proof'?

Theorem: If x is a real number, then x^2 is a positive real number.

Proof: Let P_1 be "x is positive",
P_2 be "x is negative", and
q be "x^2 is positive".

To show that $P_1 \rightarrow q$ is true, when x is positive, x^2 is positive.
Because, the product of two positive numbers $= x \cdot x = +ve$.
is positive

To show that $P_2 \rightarrow q$ is true, when x is negative, x^2 is positive.
Because, the product of two negative numbers $= (-x)(-x) = +ve$
is positive

This completes the proof.

Solution: We have missed the case when $x=0$, $x^2=0$ is not positive. So this theorem is false.

If P is "x is a real number", then we can prove results where P is the hypothesis with three cases, P_1, P_2 and P_3, where P_1 is "x is positive", P_2 is "x is negative", and P_3 is "$x=0$" be cases of the equivalence $P \rightarrow P_1 \vee P_2 \vee P_3$.

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② Exhaustive proof :-

Some theorems can be proved by examining a relatively small number of examples. Such proofs are called Exhaustive proofs, because these proofs proceed by exhausting all possibilities.

It is a special type of proof by cases where each case involves checking a single example.

Example ①: prove that $(n+1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$.

Sol: We verify the inequality $(n+1)^3 \geq 3^n$ when $n = 1, 2, 3, 84$.

For $n=1$, we have $(n+1)^3 = 2^3 = 8$ and $3^1 = 3$, so $8 \geq 3$ true.

for $n=2$, $(n+1)^3 = 3^3 = 27$ and $3^2 = 9$, so $27 \geq 9$ true.

For $n=3$, $(n+1)^3 = 4^3 = 64$ and $3^3 = 27$, so $64 \geq 27$ true.

For $n=4$, $(n+1)^3 = 5^3 = 125$ and $3^4 = 81$, so $125 \geq 81$ true.

In each of these four cases, we see that $(n+1)^3 \geq 3^n$.

∴ The method of exhaustion to prove that $(n+1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$.

③ Existence proof :-

A proof of a proposition of the form $\exists x\, P(x)$ is called an Existence proof. where P is a predicate.

[objects of a particular type exist.].

There are two types of Existence proof.

ⓐ Constructive Existence proof :-

Sometimes an existence proof of $\exists x\, P(x)$ can be given by finding an element a such that $P(a)$ is true. (or)

Sometimes an existence proof of $\exists x\, P(x)$ can be given by finding an element a, called a witness, such that $P(a)$ is true.

Example ①: prove that there is a positive integer that can be written as the sum of ~~two~~ Cubes of positive integers in two different ways.

Solution :- $1729 = 10^3 + 9^3 = 12^3 + 1^3$

[(or) $50 = 5^2 + 5^2 = 7^2 + 1^2$

(or) $65 = 8^2 + 1^2 = 4^2 + 7^2$] Sum of Squares of positive integers in two different ways.

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(b) Non-Constructive existence proof :-

We do not find an element a such that P(a) is true, but rather prove that $\exists x\, P(x)$ is true in some other way. One common method of giving this method is to use proof by contradiction and show that the negation of the existential quantification implies a contradiction.

Example(1) :- prove that there exist irrational numbers x and y such that x^y is rational.

Sol: We know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$.

If it is rational, we have two irrational numbers x & y with x^y rational, namely $x = \sqrt{2}$ and $y = \sqrt{2}$. on the otherhand, if $\sqrt{2}^{\sqrt{2}}$ is irrational, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$

so that $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = (\sqrt{2})^{\sqrt{2}\cdot\sqrt{2}} = (\sqrt{2})^2 = 2$ rational.

This proof is an example of a nonconstructive existence proof. Because we have not found irrational numbers x and y such that x^y is rational. Rather, we have shown that either the pair $x = \sqrt{2}$ and $y = \sqrt{2}$. or the pair $x = \sqrt{2}^{\sqrt{2}}$, $y = \sqrt{2}$ have the desired property, but we donot know which of these two pairs works.

(4) Uniqueness proof :-

Some theorems assert that the existence of a unique element with a particular property. (or)
Some theorems assert that there is exactly one element with this property.

To prove a statement of this type we need to show that an element with this property exists and that no other element has this property. The two parts of a Uniqueness proof are:

Existence: We show that an element x with the desired property exists.

Uniqueness: We show that if $y \neq x$, then y does not have the desired property.

Equivalently, we can show that if x & y both have the desired property, then $x = y$.

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Remark: Showing that there is a unique element x such that $P(x)$ is the same as proving the stmt

$$\exists x (P(x) \wedge \forall y (y \neq x \rightarrow \neg P(y))).$$

Example ①: Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.

Sol:

method 1: The real number $r = \frac{-b}{a}$ is a solution of $ar + b = 0$ because $a(\frac{-b}{a}) + b = 0$.

Consequently, a real number r exists for which $ar + b = 0$.

This is the existence part of the proof.

method 2: Suppose s is a real number such that $as + b = 0$. Then $ar + b = as + b$ where $r = \frac{-b}{a}$.

subtracting b from both sides, so $ar = as$.

Deviding both sides by a, but $a \neq 0$.

We see that $r = s$.

This means that if $s \neq r$, then $as + b \neq 0$

This establishes the uniqueness part of the proof.

Proof Strategies:-

Finding proofs can be a challenging (Task) bussiness.

When you take a statement to prove, you should first replace the terms by their definitions and then carefully analyze what the hypotheses and conclusion. After doing so, you can attempt to prove the result by using one of the Methods of proof.

Generally, if the stmt is a conditional stmt, you should first try with a direct proof, if this fails, you can try with an indirect proof.

If neither of these approaches works, you might try a proof by Contradiction.

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(1) Forward Reasoning :-

In a direct proof of a conditional stmt,

1) start with the premises.
2) Using these premises, together with axioms and known theorems, you can construct a proof using a sequence of steps that leads to the conclusion.

This type of reasoning called forward reasoning.
It is the most common type of reasoning used to prove relatively simple results.

Similarly, with Indirect proof of reasoning, you can
(1) start with negation of the conclusion
(2) Using sequence of steps and
(3) finally, obtain the negation of the premises.

Forward reasoning is difficult to construct to prove more complicated results, because the reasoning is needed to reach the desired conclusion may be far from obvious. In such cases it may be helpful to use backward reasoning.

Backward Reasoning :-

For proving a stmt q, we try to find a stmt p such that p is true and we can prove $p \rightarrow q$.

Example (1): prove that $\frac{x+y}{2} \geq \sqrt{xy}$ for positive real numbers x and y.

($x=4, y=6$)

(Arithmatic mean, distinct, Geometric mean.)

Sol:

$$\frac{(x+y)}{2} \geq \sqrt{xy}$$

$$\left(\frac{x+y}{2}\right)^2 \geq xy \quad \text{(square on both sides)}$$

$$\frac{(x+y)^2}{4} \geq xy$$

$$(x+y)^2 \geq 4xy$$

$$(x+y)^2 - 4xy \geq 0$$

$$(x-y)^2 \geq 0$$

Because $(x-y)^2 \geq 0$ when $x \neq y$.
This gives the backwork reasoning.

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Adapting Existing proofs:-

Existing proofs can be adapted to prove a new result. Existing proofs provide clues for new proof.

Example ①: We proved that $\sqrt{2}$ is irrational. We can adapt the proof to show that $\sqrt{3}$ is irrational?

Looking (searching) for Counterexample:-

Use of Counterexample to show that certain Statements are false.

A Stmt of the form $\forall x\, P(x)$ is false, we need only find a Counterexample, i.e an example x for which $P(x)$ is false.

Example: ① Every positive integer is the sum of the squares of two integers is false. (3 is not possible)

② "Every positive integer is the sum of the squares of three integers" is false.

Sol: successive +ve integers as a sum of three squares.

We find that $1 = 0^2+0^2+1^2$

$2^2 = 0^2+1^2+1^2$

$3 = 1^2+1^2+1^2$

$4 = 0^2+0^2+2^2$

$5 = 0^2+1^2+2^2$

$6 = 1^2+1^2+2^2$

$7 = ?$

It follows that 7 is a Counterexample.

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The Role of Open problems:- Many advances in Mathematics have been made by people trying to solve famous unsolved problems.

Fermat's Last Theorem:-

The equation $x^n + y^n = z^n$ has no solutions in integers x, y, z with $xyz \neq 0$ whenever n is an integer with $n > 2$.

Validity problem for propositional and predicate Logic:-

A formula (well formed formula (wff)) of the propositional logic is an assertion involving propositional variables by using the connectives $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ in a proper manner.

Example: $P \wedge q$ is a wff of the propositional logic.

When we assign perticular propositions to p and q, we are giving an interpretation for the formula.

Example: p: I have exam tomorrow and
q: I am going to study

$P \wedge q$: I have exam tomorrow and I am going to study.

Tautology:- A wff of the propositional logic is a tautology if the wff always takes the value true whatever interpretation is given to it. Example: $P \vee \neg P$.

Contradition:- A wff which is always false for all interpretations is called a contradiction.

Example: $P \wedge \neg P$

Contingency:- A wff which is neither a tautology nor a contradiction is called a contingency. Example: $P \rightarrow q$.

The validity problem for the propositional logic is that given a wff of propositional logic, does there exist an interpretation which will make the wff take the value true.

Considering the propositional variables as Boolean variables, this is called Boolean satisfiability problem.

A problem is decidable that an algorithm to find out if the given wff is a tautology or contingency.

The simplest way is to draw the truth table for the wff and if for all assignments it takes the value true, it is a Tautology, otherwise it is contingency.

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In the first order logic, a predicate has n arguments.
If P is a n-place predicate constant and values $c_1, c_2, \ldots c_n$ are assigned to each of the individual variables, the result is a proposition.

Suppose the domain is U. If the value of $P(c_1, c_2, \ldots, c_n)$ is true for every choice of arguments $c_1, c_2, \ldots, c_n$ selected from U, then P is said to be valid in the domain U, If the values $P(c_1, c_2, \ldots, c_n)$ is true for some (but not for all) choices of arguments selected from U, then P is said to be satisfiable in the domain U, and the values $c_1, c_2, \ldots, c_n$ which make $P(c_1, c_2, \ldots, c_n)$ true are said to satisfy P. If P is not satisfiable in the domain U, then we say P is unsatisfiable in U.

An expression involving predicate variables and quantifiers and connectives included in the proper manner is called a formula (Wff) of the first order logic.

Example: $\forall x P(x) \vee \forall x Q(x)$ is a Wff.

When we assign particular predicates to P and Q it is called an interpretation.

A Wff involving predicate variables is valid if it is true for every domain no matter how the predicate variables are interpreted.

A Wff is said to be satisfiable if there exist a domain and some interpretation of the predicate variables which makes it true.

If a Wff is not true for any domain or interpretation it is unsatisfiable.

Example: $\forall x P(x) \rightarrow \exists x P(x)$ is true for all domains and interpretation and hence a valid Wff.

$\exists x P(x) \rightarrow \exists x Q(x)$ may be true for some interpretation but may not be true for some other interpretation.

Let P(x): x is a third semester B.Tech student

Q(x): x has taken a course on Discrete Mathematics.

If Discrete Mathematics is a compulsory course for third semester students, $\exists x P(x) \rightarrow \exists x Q(x)$ will be true.

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Suppose $Q(x)$ denotes x is female student

$\exists x\, P(x) \rightarrow \exists x\, Q(x)$ may not be true, if the class has no female candidates.

If a wff is not true for any domain or interpretation, it is said to be unsatisfiable. $\forall x (P(x) \wedge \neg P(x))$ is unsatisfiable.

Given a wff of first order logic, find if it is valid or not. This is called the validity problem of the first order logic.

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